UNIT 4: Axial Compressor; Centrifugal Pump and Compressor

Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Two-dimensional pitch-line design and analysis; h-s diagram; degree of reaction; Mach number effect; performance and efficiency; three-dimensional flow; tip clearance; losses; theoretical analysis of centrifugal pump and compressor; circulation and Coriolis effects; reversed eddies; slip factor; head and efficiency; diffuser; combustion system introduction.

A. Axial compressor

1. Two-dimensional pitch-line design

Pitch-line design works at the mean radius of the blade. Velocity triangles are drawn at inlet and outlet of the rotor and stator. The usual assumptions are: axial velocity C_x is constant through the stage, and the radius at the blade root and tip is the same as at the pitch line.

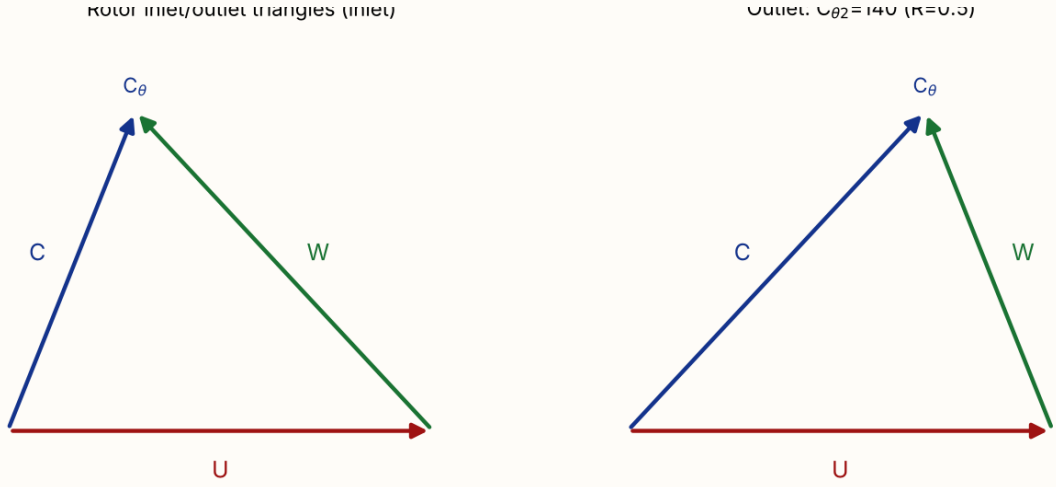
Stage work $\Delta h_0 = U\,(C_{\theta 2} - C_{\theta 1})$ **Stage loading $\psi = \Delta h_0 / U^2$**

Flow coefficient $\varphi = C_x / U$

The design process is: choose U and C_x from the required flow and pressure rise, choose the stage loading, then find the blade angles from the velocity triangles.

2. Degree of reaction

Velocity triangles for 50% reaction



The degree of reaction R is the fraction of the stage enthalpy rise that happens in the rotor:

$$\mathbf{R = \Delta h_{rotor} / \Delta h_{stage} \quad \Rightarrow \quad R = 1 - (C_{\theta 1} + C_{\theta 2}) / (2U)}$$

(constant C_x)

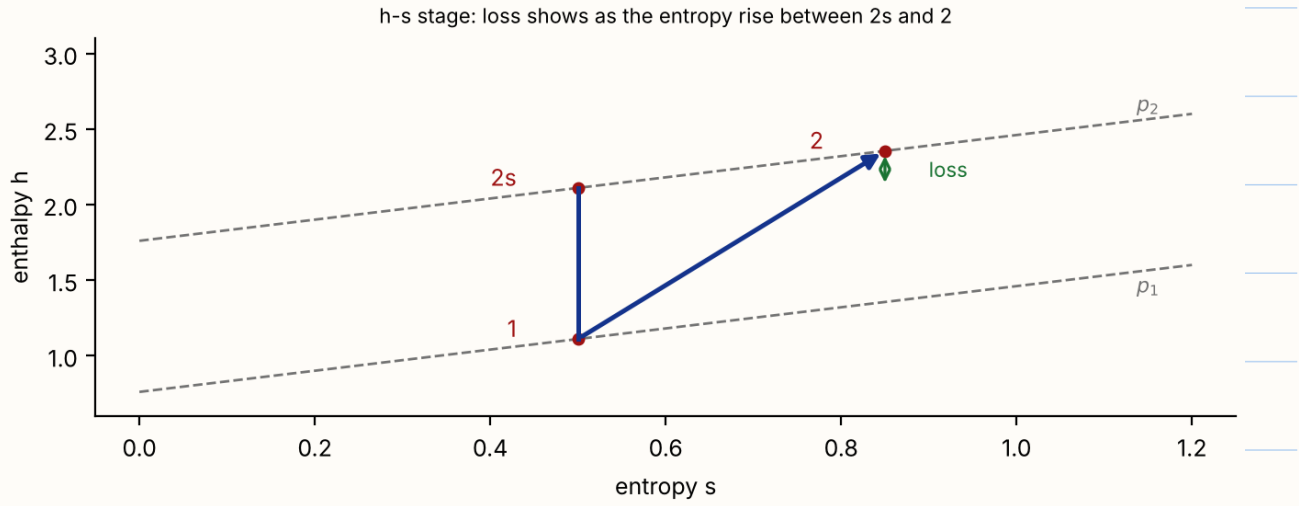
- R = 0: all the rise is in the stator (impulse-like design).
- R = 0.5: half the rise in rotor and half in stator. Symmetric velocity triangles, widely used.
- R = 1: all the rise in the rotor.

Example: *U = 200 m/s, C_x = 150 m/s, $C_{\theta 1}$ = 60 m/s. Find $C_{\theta 2}$ for 50% reaction, and the stage loading.*

Solution: R = 0.5 gives $C_{\theta 1} + C_{\theta 2} = U = 200$, so $C_{\theta 2} = 140$ m/s. $\Delta h_0 = 200 \times (140 - 60) = 16$ kJ/kg. $\psi = 16000 / 200^2 = 0.4$.

3. The h-s diagram

h-s stage with the loss gap marked



On the h-s (enthalpy-entropy) diagram, the stage is shown as a path from state 1 to state 2. The isentropic path gives Δh_{0s}, while the actual path gives Δh_0. The vertical gap between the two paths at the same pressure shows the loss. Entropy rise is caused by losses.

To sketch: draw two isobars p_{01} and p_{02}, draw the isentropic line (vertical on h-s), and then draw the actual path to the right of it. Mark the stagnation enthalpy rise Δh_0. Stage efficiency is $\eta = \Delta h_{0s} / \Delta h_0$.

4. Effect of Mach number

At the blade tip the relative velocity W is highest, because U is largest. If W reaches supersonic speed, a shock forms on the blade surface. Shocks increase losses, reduce pressure rise and can cause stall. Typical design limits are a relative tip Mach number of about 1.1 to 1.3.

Relative tip Mach number $M_{w,tip} = W_{1,tip} / a_1$

5. Diffusion factor and stall

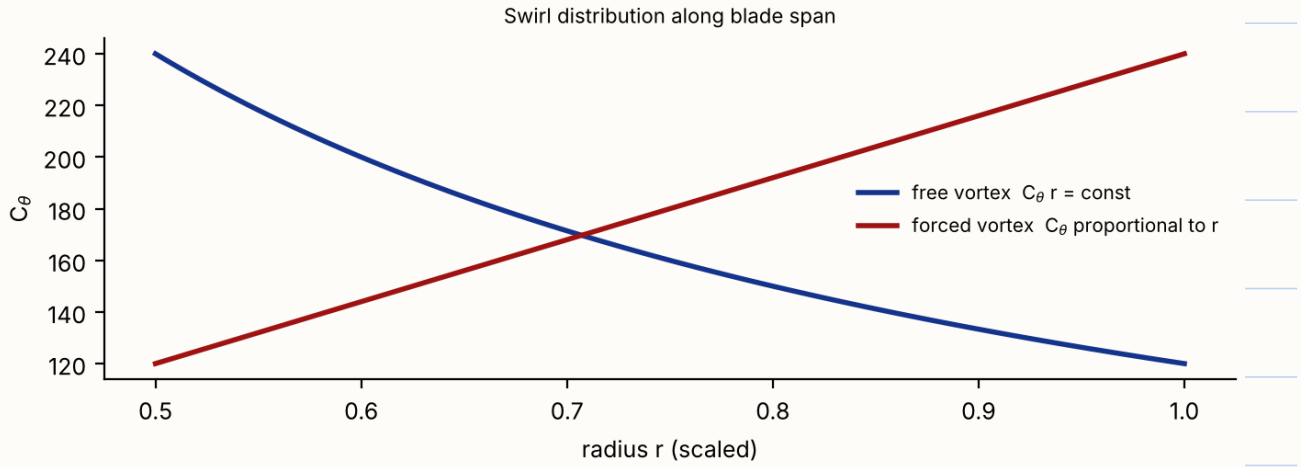
The Lieblein diffusion factor measures how much the blade row slows the relative flow. It is used to predict separation:

$DF = 1 - W_2/W_1 + \Delta C_\theta / (2\sigma W_1)$ σ = solidity = c/s

A diffusion factor below about 0.6 is generally considered safe. Higher values indicate a risk of separation and stall.

6. Three-dimensional flow and radial equilibrium

Free vortex vs forced vortex



The flow is not purely two-dimensional. Along the span there is a radial pressure gradient that balances the centrifugal force on the fluid. This is radial equilibrium:

$dp/dr = \rho C_\theta^2 / r$ (simplified radial equilibrium)

- **Free vortex design:** C_θ r = constant. Stage work is then the same at all radii, but the blade loading changes along the span.
- **Forced vortex:** C_θ proportional to r. Used in some rotors at low work.
- Designers often choose free vortex for the rotor to keep work uniform along span.

7. Tip clearance

A gap exists between the rotor tip and casing. Pressure difference across the blade drives fluid through the gap (tip leakage). The leakage flow

reduces work done on the fluid, produces a leakage vortex and increases loss. Efficiency falls roughly linearly as clearance increases. Keeping clearance small is therefore important.

8. Losses and performance

- Profile loss, secondary (endwall) loss, tip clearance loss, shock loss, and incidence loss.
- Performance is shown on a compressor map: pressure ratio against corrected mass flow, with lines of constant speed and efficiency islands.
- **Surge:** flow reverses in the compressor at low mass flow; the surge line bounds the stable range.
- **Stall:** local separation on blades, can grow into rotating stall. **Choke:** maximum flow when a passage reaches sonic speed.

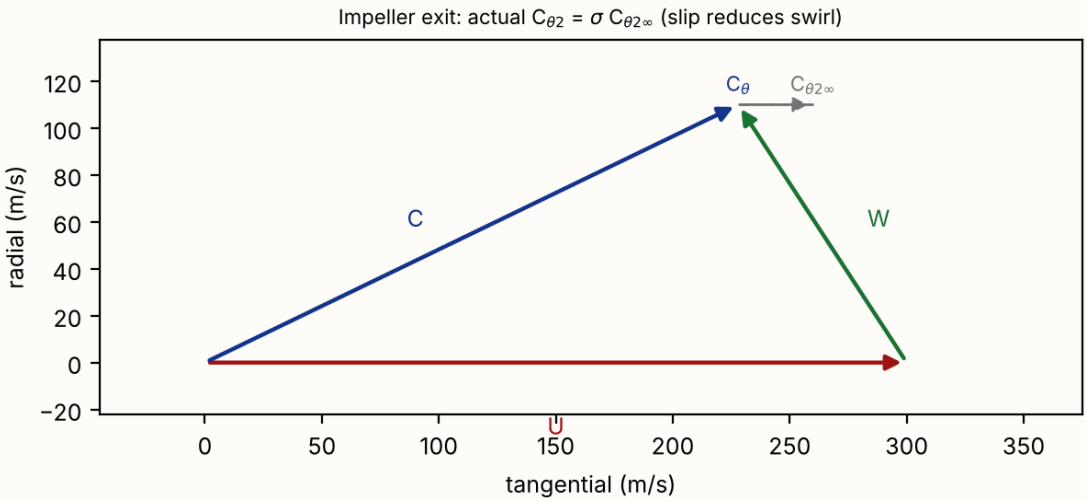
Quantity	Symbol	Meaning
Stage loading	ψ	$\Delta h_0 / U^2$, how hard the stage works
Flow coefficient	φ	C_x / U, how much flow passes
Reaction	R	Fraction of rise in rotor

Diffusion factor	DF	Predicts separation (safe below 0.6)

B. Centrifugal pump and compressor

9. Theoretical head and Euler equation

Impeller exit triangle with slip



In a centrifugal impeller, the fluid enters axially and leaves radially. The Euler head is:

$H_E = (U_2\, C_{\theta 2} - U_1\, C_{\theta 1}) / g$ with no pre-whirl ($C_{\theta 1} = 0$): $H_\infty = U_2 C_{\theta 2\infty} / g$

For an impeller with infinite blades (ideal), the exit relative flow follows the blade angle exactly. Blade angle relates the exit tangential velocity to the radial velocity:

$$C_{\theta 2\infty} = U_2 - C_{r2} \cot \beta_2 \quad \Rightarrow \quad H_\infty = (U_2^2 - U_2 C_{r2} \cot \beta_2) / g$$

10. Slip factor

With a finite number of blades, the actual exit flow turns less than the blade angle. The fluid slips. The slip factor σ is the ratio of actual to ideal tangential velocity:

$$\sigma = C_{\theta 2} / C_{\theta 2\infty} \quad \text{Stanitz: } \sigma = 1 - 0.63\pi / Z \quad \text{Wiesner: } \sigma = 1 - \sqrt{(\sin \beta_2)} / Z^{0.7}$$

Z is the number of blades. More blades give less slip. Real head is H = σ U_2 $C_{\theta 2\infty}$ / g minus losses.

Example: *A centrifugal impeller has Z = 16 and $\beta_2 = 40°$. Find the slip factor using Stanitz's formula, with $U_2 = 300$ m/s and $C_{\theta 2\infty} = 260$ m/s.*

Solution: $\sigma = 1 - 0.63\pi/16 = 1 - 0.124 = 0.876$. Actual $C_{\theta 2} = \sigma \times C_{\theta 2\infty} = 0.876 \times 260 = 228$ m/s. Ideal Euler head $H_\infty = U_2 C_{\theta 2\infty}/g = 300 \times 260 / 9.81 = 7950$ m of fluid. Slip reduces the head to about 0.876 of this, so H ≈ 6960 m of fluid.

11. Circulation, Coriolis and reversed eddies

The Coriolis effect and blade circulation produce a non-uniform relative velocity inside the impeller passage. Low-velocity, high-loss fluid collects in one part of the passage. Near the exit this distorts the flow and increases slip.

Reversed eddies: at low flow rates (below the design flow), the incidence on the inlet becomes large. Flow separates and recirculates near the inlet or near the exit. This reduces efficiency and causes unstable operation. Designers avoid operation in this region.

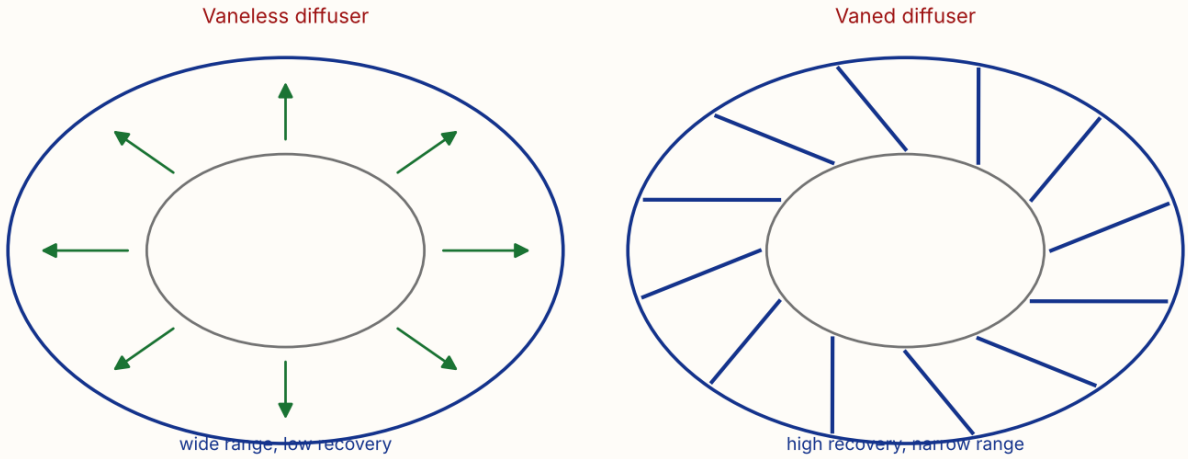
12. Head and efficiency

Hydraulic efficiency η_h = actual head / Euler head Overall η = (ρ g Q H) / shaft power

Losses in a centrifugal machine include disc friction, leakage, incidence, skin friction in the passage, and diffuser losses.

13. Diffuser

Vaneless and vaned diffusers



The diffuser recovers pressure from the high kinetic energy at the impeller exit. Two types are common:

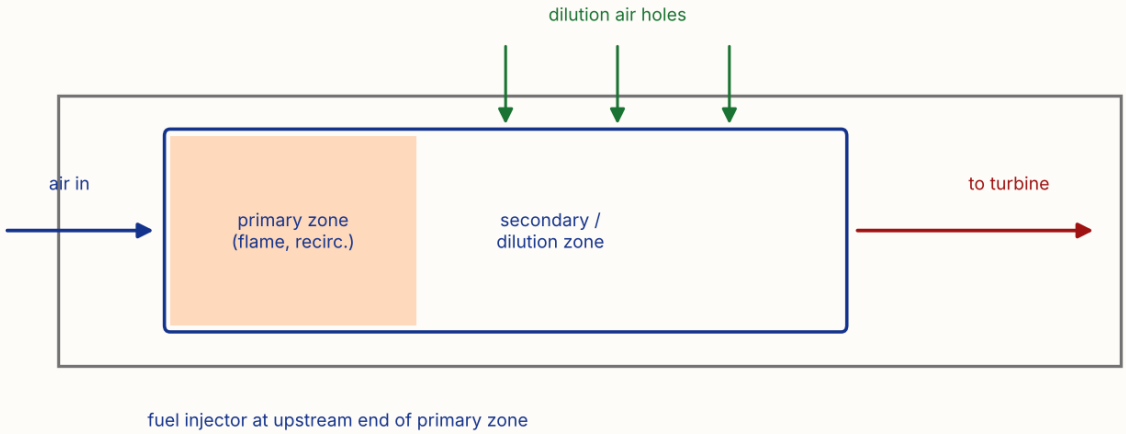
- **Vaneless diffuser:** simple, wide operating range, low pressure recovery.
- **Vaned diffuser:** higher pressure recovery, narrower operating range, more loss at off-design.

Pressure recovery coefficient $C_p = (p_2 - p_1) / (\frac{1}{2} \rho C_1^2)$

ideal: $C_{p,ideal} = 1 - (A_1/A_2)^2$

14. Introduction to the combustion system

Combustor cross-section



The combustor adds heat to compressed air by burning fuel. It must burn fuel completely, keep the pressure loss low (typically 3–6%), and produce a uniform exit temperature for the turbine. Common types are can, annular and can-annular combustors. Inside the combustor there is a primary zone (rich burning and recirculation for flame stability), a secondary zone (more air) and a dilution zone (cooling air to the turbine temperature).

Equivalence ratio $\varphi = (F/A)_{actual} / (F/A)_{stoich}$ (kerosene stoichiometric F/A ≈ 0.068, air-fuel ≈ 15)

15. Exam quick points

- Axial: $\Delta h_0 = U\Delta C_\theta$, $R = 1 - (C_{\theta 1}+C_{\theta 2})/(2U)$. Keep DF below 0.6.
- Shock at tip if $M_{w,tip}$ is too high. Tip clearance reduces efficiency.

- Centrifugal: $H = U_2 C_{\theta 2}/g$. Slip factor $\sigma < 1$, Stanitz and Wiesner.
- Reversed eddies at low flow. Vaneless diffuser for range, vaned for recovery.

16. Practice questions

Q1. Derive the expression for degree of reaction for constant axial velocity.

Q2. Explain the difference between free vortex and forced vortex designs and why the free vortex is preferred for rotors.

Q3. Explain slip factor. A centrifugal impeller has 12 blades, $\beta_2 = 35°$. Find σ using Wiesner's relation ($\sin 35° = 0.574$, $Z^{0.7} = 12^{0.7} = 5.28$).

Q4. Describe the function of the combustor primary zone and dilution zone.